VORA Research 001

How Does Orbital Altitude Affect Circular Orbital Speed and Period?

Record: VORA-RES-001 **Status:** Initial computational study — ready for internal review

Abstract

This study calculates circular orbital speed and period at five selected altitudes using the Newtonian two-body model. The results show that, within the model, circular orbital speed decreases as orbital radius increases, while orbital period increases. The outputs are mathematical calculations, not measurements from a real spacecraft.

Research question and hypothesis

Question: How do circular orbital speed and period vary with altitude above Earth's equatorial reference radius in an idealized two-body model?
Hypothesis: As altitude increases, circular orbital speed decreases and orbital period increases.

Model and assumptions

The orbit is circular; Earth is represented by a spherical-body approximation; Earth's gravitational parameter is constant; atmospheric drag, Earth's nonspherical gravity field, and third-body perturbations are neglected. Altitude is measured above the adopted equatorial radius.

Constants and equations

Earth gravitational parameter: $\mu = 398600.4\ km^3/s^2$, rounded from the JPL value $398600.435507\ km^3/s^2$ [1]. Earth equatorial radius: $R_\blacksquare = 6378.137\ km$ [2].

$r = R_\blacksquare + h$
$v = \sqrt{(\mu / r)}$
$T = 2\pi \sqrt{(r^3 / \mu)}$

Results

Altitude (km)	Radius (km)	Speed (km/s)	Period (min)	Period (h)
200	6578.137	7.784	88.49	1.4749
400	6778.137	7.669	92.56	1.5427
800	7178.137	7.452	100.87	1.6812
2000	8378.137	6.898	127.20	2.1200
35786	42164.137	3.075	1436.07	23.9344

Interpretation

The calculated circular speed decreases with increasing orbital radius because v varies with $r_\blacksquare^{-1/2}$. The orbital period increases because T varies with $r^{3/2}$. The five cases support the hypothesis within the assumptions of the model. At 35,786 km altitude, the calculated period is approximately 23.934 hours, close to Earth's sidereal rotation period. A

geostationary orbit must also be circular, equatorial, and prograde.

Limitations

This is a simplified analytical calculation, not a numerical propagation of a real spacecraft and not an experimental measurement. Real orbits can be affected by atmospheric drag, Earth's oblateness, lunar and solar gravity, solar radiation pressure, and nonzero eccentricity. The study does not assess launch requirements, station-keeping, satellite lifetime, or mission suitability.

Reproducibility

Run `python vora_res001_calculate.py` with Python 3. The script writes `vora_res001_results.csv`. The plots in the research package show selected calculated sample points, not observational data.

References

[1] NASA/JPL Solar System Dynamics, “Astrodynamic Parameters.” https://ssd.jpl.nasa.gov/astro_par.html

[2] NASA Goddard Space Flight Center, NSSDC, “Earth Fact Sheet.” https://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html

[3] D. Chitwood, “Appendix B: Orbital Mechanics Equations,” NASA Technical Reports Server. https://ntrs.nasa.gov/api/citations/19950006116/downloads/19950006116.pdf

Publication note

This is an introductory computational study, not a claim of novel scientific discovery. Before public release, check source links, code output, graph labels, displayed rounding, and the final report.